Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory (`https://assumptionsofphysics.org/resources/briefs`) were checked in Lean, where the formal version differs from the brief, and any problems found in the brief.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Equivalence of sup-norm, L^1, and L^2 seminorms on the Schwartz space

Files.

- File: `briefs/SchwartzNorms/SchwartzNorms.lean`
- Library: `SchwartzNorms`
- Imports: `SchwartzMomentProblem` (for the moments in eq. (10)) and three new `ForMathlib` files:
  `ForMathlib/Analysis/LocallyConvex/WithSeminorms.lean` (bounds between seminorm families compose),
  `ForMathlib/Analysis/SpecialFunctions/ImproperIntegrals.lean` ($\int (1+x^2)^{-2} = \pi/2$, $\int_{|x|>1} x^{-2} = 2$),
  `ForMathlib/MeasureTheory/Integral/IntegralEqImproper.lean` ($\|g(x)\| \le \|g'\|_{L^1}$ for integrable g, g')

Setting. For $f : \mathbb{R} \to \mathbb{C}$ and $g = x^\alpha \partial^\beta f$ (`weighted`), the seminorms (1)–(3) are `supNorm`, `l1Norm` and `l2Norm`, with values in $[0, \infty]$ (`eLpNorm` for L^1 and L^2). Smoothness is `ContDiff` of order ∞. On $\mathcal{S}(\mathbb{R})$, the L^1 and L^2 seminorms are real seminorms `l1Seminorm`, `l2Seminorm`, collected in the families `l1Family`, `l2Family`; the sup family is Mathlib's `schwartzSeminormFamily`.

Results.

Brief	Lean
eq. (6), sup controls L^2	`l2Norm_sq_le`
eq. (7), L^2 controls L^1	`l1Norm_le`
eq. (9), L^1 controls sup	`supNorm_le` (with $\|f\|^{L^1}_{\alpha,\beta} < \infty$); `not_supNorm_le_const` (the hypothesis is needed)
Theorem 5, (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii)	`tfae_finite`
definition (4) is Mathlib's $\mathcal{S}(\mathbb{R})$	`exists_schwartzMap_iff`, `supNorm_eq_seminorm`
Step 4, same topology	`withSeminorms_l1_l2`, from `isBounded_sup_l2`, `isBounded_l2_l1`, `isBounded_l1_sup` and `WithSeminorms.congr_of_cycle`
eq. (10)	`enorm_moment_le` (with the factor $\|\hbar\|^\beta$)

Differences from the brief.

- The seminorms take values in $[0, \infty]$, so that Theorem 5 can be stated for every smooth f. On $\mathcal{S}(\mathbb{R})$ they are finite and give real seminorms.
- "Generate the same locally convex topology" is stated as `WithSeminorms`: each family induces Mathlib's topology on $\mathcal{S}(\mathbb{R})$. Step 4's "this relation is transitive" is `Seminorm.IsBounded.comp`, new in `ForMathlib`.
- In Step 3, $g(x) = \int_{-\infty}^{x} g'$ uses Mathlib's improper FTC and the fact that an integrable g with integrable g' tends to 0 at $-\infty$.
- The converse in the remark after eq. (10) (finite expectation values imply membership in $\mathcal{S}(\mathbb{R})$) and the remark on general L^p are not formalized.

Problems found in the brief.

- Eq. (9) needs $\|f\|^{L^1}_{\alpha,\beta} < \infty$, so that $g \to 0$ at $-\infty$. The brief has it from (ii), but (9) is stated as an inequality between seminorms, and it is false without it: for $f = 1$, $\alpha = \beta = 0$, the left side is 1 and the right side is 0.
- In eq. (7) the coefficient of $\|f\|^{L^2}_{\alpha+1,\beta}$ is 2; the argument given ($\sqrt{a+b+c} \le \sqrt{a} + \sqrt{b} + \sqrt{c}$ with $b = 2(\|f\|^{L^2}_{\alpha+1,\beta})^2$) gives $\sqrt{2}$. The stated bound is still true, and is the one formalized.
- Eq. (10) omits the factor $|\hbar|^\beta$ from $P^\beta = (-i\hbar\partial)^\beta$.